Second Year- Fourth Semester

Subject Code	Subject Name	Scheme of Teaching				Scheme of Examination		
						Theory		Practical (Internal)
		L-T-P	Contact hrs./week	Credits	Category	Internal Ass.	Univ. Exam	
HSMC-401	Elective- I (from Humanities and Social Sciences)	3-0-0	3	3	Value Added Course	50	50	-
ASM-401	Discrete Structures	3-1-0	4	4	BS	50	50	-
PCIT401	Microprocessor & Assembly Language Programming	3-1-0	4	4	PC	50	50	-
PCIT451	Microprocessor & Assembly Language Programming(P)	0-0-3	3	1	PC	-	-	50
PCIT402	Computer Networks	3-1-0	4	4	PC	50	50	-
PCIT452	Computer Networks(P)	0-0-3	3	1	PC	-	-	50
PCIT403	Operating System	3-1-0	4	4	PC	50	50	-
PCIT453	Operating System(P)	0-0-3	3	1	PC	-	-	50
Total		**15-4-09**	**28**	**22**		**250**	**250**	**150**

Total Marks: 650 **Total Credits: 22**

Elective-I (from Humanities and Social Sciences**)**

- HSMC-401a Economics
- HSMC-401b Introduction to Psychology
- HSMC-401c Sociology
- HSMC-401d Russian Language

SYLLABUS FOR B.E. (I.T.) FOURTH SEMESTER

COURSE INFORMATION SHEET

Course Code	**HSMC-401a**
Course Title	**Economics (Theory)**
Type of Course	**Value Added Course**
L T P	**3 0 0**
Credits	**03**
Total Lectures	**45**
Course Assessment Methods: End Semester Assessment (University Exam.) Continuous Assessment (Sessional)	 **50** **50**
Course Prerequisites	**Nil**
Course Objectives	I. To make students understand how society manages its scarce resources for achieving maximum satisfaction. II. To make students learn about economic aspects related to a consumer, firm, market and economy.
Course Outcomes	After completion of this course, the students will be able to: I. Apply engineering knowledge to maximize profit, satisfaction and welfare. II. Identify the forces that affect the economy. III. Learn entrepreneurial skills and analyze the concepts of demand and supply. IV. Develop analytical skills in students to understand different markets.

SYLLABUS

Note: The examiner shall set seven questions of 10 marks each. First question has to be compulsory, having sections covering the whole syllabus. Three questions have to be set from Section A and three questions from Section t B of the syllabus. Candidate is required to attempt at least two questions from each section. All the course outcomes must be covered by the question paper.

SECTION-A	**Hours**
Introduction to Economics Nature of Economics, Economic Thoughts, Economic Activities, Relationship of Economics with other Social Sciences and Engineering.	**(06)**
Theory of Consumer Behaviour Demand: Types, Law of Demand, Determinants of Demand and Change in Demand Elasticity of Demand: Nature, Degrees, Types, Measurement and Factors Affecting Elasticity of Demand and its Application. Laws of Consumption: Concept and Applicability of Law of Diminishing Marginal Utility and Law of Equi-Marginal Utility.	**(12)**
Theory of Production and Cost Cost: Types of Costs, Production: Law of Variable Proportion, Returns to Factor and Returns to Scale, Economies and Diseconomies of Scale.	**(06)**

SECTION-B	
Theory of Market Nature and Relevance of Perfect Competition, Monopoly and Monopolistic Competition.	**(08)**
Basic Concepts of Macro Economics National Income: Concept and Measurement, Determination of Equilibrium of Income Inflation: Concept, Causes and Effect of Inflation, Measures to Control Inflation.	**(09)**
Project Presentations	**(04)**

RECOMMENDED BOOKS

S. No.	NAME	AUTHOR(S)	PUBLISHER
1	Modern Economics	H. L. Ahuja	S. Chand & Co. Ltd
2	Economics For Engineers	M.L. Gupta. & S.P. Gupta	ESS PEE Publications
3	Business Economics	H.L. Ahuja	S. Chand & Co. Ltd
4	Macro Economic Theory	M.L. Jhingan	Konark Publisher Pvt. Ltd
5	Principles of Microeconomics	J. Stiglitz& Carl E Walsh	W.W. Norton & Company
6	Principles of Economics	Mankiw N Gregory	Cengage Learning
7	Course in Micro Economics Theory	A. Kreps	Prentice Hall
8	Economics	Samuelson A. Paul &Nordhaus D William	Tata McGraw Hill
9	Microeconomics	H. Gravelle& R. Reiss	Pearson Education
10	Macro Economics: Theory and Practice	H. L. Ahuja	S. Chand & Co. Ltd.
11	Economics for engineers	T.R Jain, M.L Grover & V.K Ohei	V.K Publications

COURSE INFROMATION SHEET

Course Code	**ASM-401**
Course Title	**Discrete Structures (Theory)**
Type of Course	**BS**
L T P	**3 1 0**
Credits	**04**
Total Lectures	**45**
Course Assessment Methods End Semester Assessment (University Exam.) Continuous Assessment (Sessional)	 **50** **50**
Course Prerequisites	**Nil**
Course Objectives	I. To get familiar and understand the fundamental notions in discrete mathematics. II. To introduce the knowledge of core mathematical foundation of computer science. III. Be exposed to concepts and properties of algebraic structures such as semi groups, monoids and groups. IV. Be aware of the counting principles. V. To introduce the basic properties of graphs and trees and model simple applications.
Course Outcomes	After completion of this course, the students will be able to: I. Get familiar and understand the fundamental notions in discrete mathematics. II. Acquire the knowledge of core mathematical foundation of computer science. III. Aware of the counting principles, basic properties of graph, trees and model simple applications. IV. Exposed to concepts and properties of algebraic structures such as semi groups, monoids and groups.

SYLLABUS

Note: The examiner shall set seven questions of 10 marks each. First question has to be compulsory, having sections covering the whole syllabus. Three questions have to be set from Section A and three questions from Section B of the syllabus. Candidate is required to attempt at least two questions from each section. All the course outcomes must be covered by the question paper.

SECTION-A	**Hours**

Sets, Relations and Functions Definition of sets, product sets and sectionitions, Relations and digraphs, matrix of a relation, paths in relations and digraphs, equivalence relations and sectionitions, operations on relations, transitive closure and warshall's algorithm.(Scope as in Chapter 4, Sections 4.1 – 4.7 of Reference 2). Functions, One-to-one and onto functions, Special functions. The pigeon hole principle. Function composition and inverse functions (Scope as in Chapter 5, Sections 5.1 – 5.6 of Reference 1). Sectionially ordered sets; Extremal elements of Sectionially ordered sets, Lattices, Linearly ordered sets. (Scope as in Chapter 6, Sections 6.1 – 6.3 of Reference 1).	**(14)**
Fundamentals of Logic Basic connectives and truth tables, Logical equivalence, The laws of logic, Logical implication, Rules of Inference, Use of Quantifiers, Definitions and Proofs of Theorems (Scope as in Chapter 2, Sections 2.1 – 2.5 of Reference 1).	**(8)**
SECTION-B	
Principles of Counting Rule of Sum and Product, Permutations, Combinations, Combinations with repetition (Scope as in Chapter 1, Sections 1.1 – 1.4 of Reference 1). The principle of Inclusion and Exclusion, Generalizations, Derangements (Scope as in Chapter 8, Sections 8.1 – 8.3 of Reference 1). Generating Functions: Definitions and Examples: Calculation Techniques, Sectionitions of Integers, The exponential generating function, The summation operator (Scope as in Chapter 9, Sections 9.1 – 9.5 of Reference 1). Recurrence relations: The first order linear recurrence relation, The second order linear homogeneous recurrence relation with constant coefficients, The non homogeneous recurrence relation, The method of generating functions (Scope as in Chapter 10, Sections 10.1 – 10.4 of Reference 1).	**(9)**
Graph Theory Definitions and examples, Subgraphs, Complements and Graph Isomorphism, Vertex degree: Euler trails and circuits, Planar Graphs, Hamilton Paths and Cycles, Graph colouring and Chromatic polynomials (Scope as in Chapter 11, Sections 11.1 – 11.6 of Reference 1).	**(5)**
Groups Theory Definition and elementary properties of groups, subgroups, Homomorphism, Isomorphism and Cyclic groups, Cosets and Lagrange's Theorem (Scope as in Chapter 16, Sections 16.1 – 16.3 of Reference 1). Introduction to Rings and Fields (definitions, examples and basic properties) (Scope as in Chapter 14, Sections 14.1-14.2 of Reference 1).	**(9)**

RECOMMENDED BOOKS

S. No.	NAME	AUTHOR(S)	PUBLISHER
1	Discrete and Combinatorial Mathematics	Ralph P. Grimaldi	Pearson Education, 4th Edition
2	Discrete Mathematical Structures	B. Kolman, R. C. Busby and S. C. Ross	Pearson Education, 5th Edition
3	Elements of Discrete Mathematics	C.L.Liu, D P Mohapatra	Tata McGraw Hill
4	Discrete Mathematics for Computer Scientists and Mathematicians	J. L. Mott, A. Kandel, T. P. Baker.	Prentice-Hall of India, 2nd Edition
5	Discrete Mathematics and applications	K.H.Rosen	Tata McGraw Hill
6	Discrete Mathematics	S. Lipschutz, M. Lipson	Schaum's Outlines, Tata McGraw-Hill, 2nd Edition

COURSE INFORMATION SHEET

Course Code	**PCIT 401**
Course Title	**Microprocessor & Assembly Language Programming (Theory)**
Type of Course	**PC**
L T P	**3 1 3**
Credits	**04**
Total Lectures	**45**
Course Assessment Methods: End Semester Assessment (University Exam.) Continuous Assessment (Sessional)	 **50** **50**
Course Prerequisites	, Computer Architecture and Organization.
Course Objectives	To understand and apply the concepts of 8085 Microprocessor so as to prepare the graduates to write assembly language programs for solving various problems.
Course Outcomes	After completion of this course, the students will be able to: I. Recall the concepts of number system, digital circuits, data buses, memory, registers, ports. II. Understand the architecture of 8085 and its interfacing with Memory and peripheral I/O devices. III. Apply the concepts of microprocessor to write assembly language programs using 8085 programming instructions. IV. Analyze the problems, time delays caused and breakdown the problem into subroutines. V. Synthesis and develop solution in assembly language by employing the concepts of stacks, subroutines, interrupts and various Programmable Peripheral devices. VI. Evaluate and assess the output of assembly language program.

SYLLABUS

Note: The examiner shall set seven questions of 10 marks each. First question has to be compulsory, having sections covering the whole syllabus. Three questions have to be set from Section A and three questions from Section B of the syllabus. Candidate is required to attempt at least two questions from each section. All the course outcomes must be covered by the question paper.

SECTION-A	**CO(s)**
Microprocessor Architecture and Microcomputer Systems Microprocessor Architecture, The 8085 MPU: Block Diagram, Pin Diagram, Address/Data Buses, Concept of de-multiplexing of Buses, Control and status signals, Registers, Ports, Flags, Instruction Decoding and Execution, memory Interfacing. **(06)**	**1,2**

Interfacing I/O Devices Basic Interfacing Concepts, Interfacing Output Displays, Interfacing Input Devices, Memory-Mapped I/O. **(06)**	**2**
Programming the 8085 Introduction to 8085 Assembly Language Programming, The 8085 Programming Model, Instruction Classification, Instruction Format. Data Transfer (Copy) Operations, Arithmetic Operations, Logic Operations, Branch Operations, Writing Assembly Language Programs. **(07)**	**3,6**
Programming Techniques with Additional Instructions Programming Techniques Looping, Counting and Indexing, Additional Data Transfer and 16-Bit Arithmetic Instructions, Arithmetic Operations Related to Memory, Logic Operations. **(6)**	**3,6**
SECTION-B	
Counters and Time Delays Counters and Time Delays, Hexadecimal Counter, Modulo Ten, Counter, Generating Pulse Waveforms, Debugging Counter and Time-Delay Programs. **(06)**	**4,6**
Stack and Subroutines Stack, Subroutine, Conditional Call and Return Instructions. **(04)**	**5**
Interrupts The 8085 Interrupt, 8085 Vectored interrupts. **(03)**	**5**
General –Purpose Programmable Peripheral Devices Block Diagram, Working and Control word of: The 8255A Programmable Peripheral Interface, The 8259 A Programmable Interrupt Controller, Programmable communications interface 8251. **(07)**	**5**

RECOMMENDED BOOKS

S. No.	NAME	AUTHOR(S)	PUBLISHER
1	Microprocessor Architecture, Programming and Applications with the 8085	Ramesh S.Gaonkar	PHI
2	Advanced Microprocessors & Interfacing	Badri Ram	Tata McGraw Hill
3	Microprocessor Principles and Applications	Charles M.Gilmore	Tata McGraw Hill
4	Microprocessors and Interfacing programming and Hardware	Douglas V. Hall	Tata McGraw Hill

COURSE INFORMATION SHEET

Course Code	**PCIT 451**
Course Title	**Microprocessor & Assembly Language Programming (Practical)**
Type of Course	**Core**
Credits	**1**
Course Assessment Methods: End Semester Assessment (University Exam.) Continuous Assessment (Practical)	 **00** **50**
Course Prerequisites	Basic Electrical and Electronics Engineering, Computer Architecture and Organization.
Course Objectives	To develop, key-in, test and troubleshoot the assembly language program and machine level program on 8085 kits.

SYLLABUS

- Familiarization of 8085 kits.
- Application of assembly language using 8085 instructions set to develop various programs.

COURSE INFORMATION SHEET

Course Code	**PCIT 402**
Course Title	**Computer Networks (Theory)**
Type of Course	**PC**
L T P	**3 1 2**
Credits	**04**
Total Lectures	**45**
Course Assessment Methods: End Semester Assessment (University Exam.) Continuous Assessment (Sessional)	 **50** **50**
Course Prerequisites	Basic Electrical and Electronics Engineering
Course Objectives	This course is to provide students with an overview of the concepts of data communication and computer networks. The main course objectives are: I. To Familiarize with the basic taxonomy,terminology and functioning of computer networks. II. Building an understanding of various existing protocols and functions of layer
Course Outcomes	After completion of this course, the students will be able to: I. To understand the basic concepts of computer network including various reference models and protocols. II. To familiarize with various transmission media used in computer networks. III. To recognize the concept of different techniques of flow control and error control during data transmission and illustrate various protocols of data link layer and MAC sub-layer. IV. To learn the basic functions of network layer and apply the concepts of routing and IP addressing to solve problems. V. To recognize the various functions of transport layer compare and contrast the data link layer and transport layer. VI. To understand the concepts of various protocols of application layer

SYLLABUS

Note: The examiner shall set seven questions of 10 marks each. First question has to be compulsory, having sections covering the whole syllabus. Three questions have to be set from Section A and three questions from Section B of the syllabus. Candidate is required to attempt at least two questions from each section. All the course outcomes must be covered by the question paper.

SECTION-A	**CO(s)**

Introduction Basic concepts of computer networks,; Network Hardware: LAN, MAN, WAN, Wireless networks, Internet; Network Software: Layer, Protocols, interfaces and services; Reference Model: OSI/TCP/IP and their comparison. **(08)**	**1**
Physical Layer Multiplexing, Line coding techniques, Transmission media: Magnetic, Twisted pair, coaxial cable, fiber optics, wireless transmission (radio, microwave, infrared, light wave). Switching: Circuit Switching & Packet Switching. Cellular radio and communication satellites. **(08)**	**2**
Data Link Layer Framing, Error control: Error correction & Detection, sliding window protocols (one bit, Go back n, selective repeat), Medium Access Sub layer: Channel Allocation, MAC protocols -ALOHA, CSMA protocols, Collision free protocols, IEEE 802.3, 802.4, 802.5 standards and their comparison. **(09)**	**3**
SECTION-B	
Network Layer Design issues, routing algorithms (shortest path, flooding, flow based, distance vector, hierarchical, broadcast, multicast).Congestion control algorithms (Leaky bucket, Token bucket, Choke, Packet, Load shedding), IPV4, IP addressing, IPV6. **(09)**	**4**
Transport Layer Addressing, establishing and releasing connection, flow control & buffering, multiplexing, crash recovery, Internet Transport protocol (TCP and UDP). **(06)**	**5**
Application Layer Network Security; Domain Name System; Simple Network Management Protocol; Electronic Mail. **(05)**	**6**

RECOMMENDED BOOKS

S. No.	NAME	AUTHOR(S)	PUBLISHER
1	Computer Networks, 4th Edition	Andrew S. Tanenbaum	Prentice Hall of India
2	Data and Computer Communications	William Stallings	Prentice Hall of India
3	Data Communication and Networking	Behrouz A Forouzan	Tata McGraw Hill
4	Design & Analysis of Computer Communication Networks	Vijay Ahuja	McGraw Hill
5	Data Communications and Networks	Douglas E. Coomer	Prentice Hall of India

COURSE INFORMATION SHEET

Course Code	**PCIT 452**
Course Title	**Computer Networks(Practical)**
Type of Course	**PC**
Credits	**1**
Course Assessment Methods: End Semester Assessment (University Exam.) Continuous Assessment (Practical)	 **00** **50**
Course Prerequisites	Introduction to Information Technology, Basic Information Theory and Communication.
Course Objectives	This course is to provide students with an overview of the concepts of data communication and computer networks. The main course objectives are: 1. Familiarize the student with the basic taxonomy, terminology and functioning of computer networks. 2. Building an understanding of various existing protocols for data communication in computer networks.

SYLLABUS

1.To Study different type of network cables.
2.To implement the cross-wired cable and straigt cable using clamping tool.
3.To study various network devices in detail.
4.To study IP address subnetting.
5.To connect the computers in LAN.
6.To Study basic network command and networks communication commands.
7.To configure a network topology.
8.Cisco packet tracer can be used to configure networks.

COURSE INFORMATION SHEET

Course Code	**PCIT 403**
Course Title	**Operating System (Theory)**
Type of Course	**PC**
L T P	**3 1 3**
Credits	**04**
Total Lectures	**45**
Course Assessment Methods: End Semester Assessment (University Exam.) Continuous Assessment (Sessional)	 **50** **50**
Course Prerequisites	**-**
Course Objectives	I. To study and understand main components of operating system, their working, and operations performed by operating system. II. To provide students knowledge on: resource management provided by operating systems, concepts and theories of operating systems, implementation issues of operating systems. III. To be able to understand description of multiprocessor and distributed operating system and different operating system and compare their features.
Course Outcomes	After completion of this course, the students will be able to: I. Describe the important computer system resources and role of operating system in their management policies and algorithms. II. Understand the process management policies and scheduling of processes by CPU and Evaluate the requirement for process synchronization and coordination handled by operating system. III. Describe and analyse the memory management and its allocation policies. IV. Identify use and evaluate the storage management policies with respect to different storage management technologies. V. Analyse and Identify the need to create the special purpose operating system and services provided by Distributed operating system. VI. Compare and contrast the services provided by UNIX operating system.
SYLLABUS	
Note: The examiner shall set seven questions of 10 marks each. First question has to be compulsory, having sections covering the whole syllabus. Three questions have to be set from Section A and three questions from Section B of the syllabus. Candidate is required to attempt at least two questions from each section. All the course outcomes must be covered by the question paper.	

SECTION-A	CO(s)
Basic Functions and Concepts of Operating Systems Concept of an operating systems, batch system, Multi-programmed, Time sharing, Personal Computer System, Parallel system, Real time system, General system Architecture. **(05)**	**1**
Features and Objectives of Operating Systems System components, operating system services, System calls, System Programs, System Structure, System design and implementation. Concept of process, process states, process state transition, process control block, operations of processes, concurrent processes, deadlocks, scheduling algorithms, scheduling criteria, Process Synchronization. **(11)**	**2**
Memory Management Logical and physical address space, storage allocation and management techniques, swapping, concepts of multi programming, paging, segmentation, virtual storage management strategies, Demand Paging, Page Replacement Algorithms, and Thrashing. **(06)**	**3**
SECTION-B	
Information Management File concept, Access method, Directory structure, Protection File system structure, Allocation methods, Free space management, Directory implementation, Disk structure, Disk Scheduling, Disk management, Swap space management. **(06)**	**4**
Distributed-System Structures Network operating system, Distributed operating systems, Remote services, Robustness, Design Issues. **(06)**	**5**
Distributed file systems and Distributed Coordination Naming and Transparency, Remote file Access, Stateful versus stateless service, File replication, Event ordering, Mutual Exclusion, Atomicity, Concurrency control, Deadlock Handling, Election Algorithms, Reaching Agreement. **(06)**	**5**
Case Studies: Unix O.S. Architecture, Operating system services, user perspective, representation of files in Unix system processes and their structure, Input-output system, Memory management, Unix shell, history and evolution of Unix system. **(05)**	**6**

RECOMMENDED BOOKS

S. No.	NAME	AUTHOR(S)	PUBLISHER
1	Operating Systems, 5th Edition	Galvin & Silberschatz	Addison Wesley Publishing Ltd
2	An Introduction to Operating System, 3rd Edition	Harvey M. Deitel	Narosa Publishing House
3	Operating Systems: Design and implementation, 3rd Edition	Andrew S. Tanenbaum	PHI
4	Operating system, 5th Edition	Millan Milankovic	McGraw Hill

COURSE INFORMATION SHEET

Course Code	**PCIT 453**
Course Title	**Operating System (Practical)**
Type of Course	**PC**
Credits	**1**
Course Assessment Methods: End Semester Assessment (University Exam.) Continuous Assessment (Practical)	 **00** **50**
Course Prerequisites	Programming Fundamental. Object Oriented Programming using C++
Course Objectives	I. To teach students about various operating systems including Windows, and UNIX. II. To be able to students learn about systems configuration and administration. Students learn, explore and practice technologies related to UNIX.

SYLLABUS

List of Practicals:

1. Implement various CPU scheduling algorithms.
2. Write program to implement banker's algorithm for deadlock prevention.
3. Write programs to implement Page replacement algorithms.
4. Write an algorithm and program to implement Disc scheduling.
5. Installation of the Linux operating system
6. Using basic commands-man, who, more, pipe, finger, cat, redirect, ls, cp, mv, rm.Working with directory and plain files-pwd, cd, mkdir, rmdir, lp, wc, date, cal, sort, diff, uniq and grep commands.
7. Using miscellaneous commands-head, tail, cut, copy, paste, spell, find and bc.
8. Working with shell scripts under Korn Shell and using shell variables, print, chmod and calendar commands.
9. Using quotes, relational operators, command substitution, arithmetic functions, shell control statements such as for-in, if-then-elseif-else, while,case,date and script.
10. Working under the Bourne shell-shell scripts, control statements such as test, for, for in, if-then-else-fi, -if-then-elif-fi, while,until, case, relational operators and expressions.